Software Engineering – Sample Study Notes

CampusAI demonstration material. These are sample notes; verify and expand them using your course syllabus.

What is Software Engineering?

- Software engineering is a systematic approach to developing, testing, maintaining, and improving software.
- It aims to create reliable software within time and budget.

SDLC

- SDLC means Software Development Life Cycle.
- Typical phases: planning, requirements, design, development, testing, deployment, and maintenance.

Software Process Models

- Waterfall completes phases in sequence.
- Iterative development improves software through repeated cycles.
- Agile develops in small increments with frequent feedback.

Software Testing

- Testing checks whether software works as expected and helps find defects.
- Unit testing checks a component; integration testing checks combined components; system testing checks the complete system.

Requirements

- Functional requirements describe what the system should do.
- Non-functional requirements describe quality attributes such as performance, security, and usability.